

DSC 140A

Probabilistic Modeling & Machine Learning

Lecture 15 | Part 1

Recap

Applying the Bayes Classifier

- Predict the class y which maximizes:

$$p_X(\vec{X} = \vec{x} \mid Y = y) \mathbb{P}(Y = y)$$

- We must **estimate** the density, p_X .
- Two approaches:
 1. Non-parametric (e.g., histograms)
 2. Parametric (e.g., fit Gaussian with MLE)

Curse of Dimensionality

- ▶ In practice, we have many features.
- ▶ This means $p_X(\vec{X} = \vec{x} \mid Y = y)$ is **high dimensional**.
- ▶ Non-parametric estimators do not do well in high dimensions due to the **curse of dimensionality**:
 - ▶ Data required grows exponentially with number of features.

Responses

- ▶ Parametric density estimation can fare better.
- ▶ However, it too can suffer from the curse.
- ▶ **Today**, a different approach: assume **conditional independence**.

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Probabilistic Modeling & Machine Learning

Lecture 15 | Part 2

What is Conditional Independence?

Remember: Independence

- Events A and B are **independent** if

$$\mathbb{P}(A, B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

- Equivalently, A and B are independent if¹

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A, B)}{\mathbb{P}(B)} \stackrel{\text{if } A, B \text{ ind.}}{=} \frac{\mathbb{P}(A) \cancel{\mathbb{P}(B)}}{\cancel{\mathbb{P}(B)}} = \mathbb{P}(A)$$

¹or $\mathbb{P}(B) = 0$

Informally

- ▶ A and B are **independent** if learning B does not influence your belief that A happens.

$$P(A, B) = \frac{4}{52} = \frac{1}{13}$$

Example

$$P(A) = 1/4$$

$$P(B) = \frac{16}{52} = \frac{8}{26} = \frac{4}{13}$$

You draw one card from a deck of 52 cards. A is the event that the card is a heart, B is the event that the card is a face card (J, Q, K, A). Are these independent?

$$P(A, B) = P(A) P(B) \quad \text{Yes, indep!}$$

♥: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♦: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♣: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♠: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

$$P(A, B) = \frac{4}{51}$$

Example

$$P(A) = \frac{13}{51}$$

$$P(B) = \frac{15}{51}$$

We've lost the King of Clubs! You draw one card from this deck of 51 cards. A is the event that the card is a heart, B is the event that the card is a face card (J, Q, K, A). Are these independent? *No!*

$$P(A, B) \neq P(A)P(B)$$

$$P(A|B) \stackrel{?}{=} P(A)$$

$$P(B|A) \stackrel{?}{=} P(B)$$

♥: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♦: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

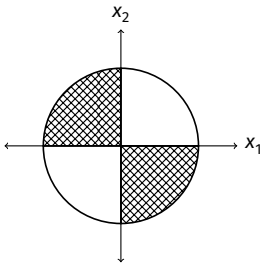
♣: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, A

♠: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

Exercise

Suppose a dart throw is uniformly distributed on the dartboard below. Are X_1 and X_2 independent?

No, dependent!



In the Real World...

- ▶ ...true independence is rare.
- ▶ Example, survivors of the titanic:

PassengerID	Survived	Pclass	Sex	Age	Fare	Embarked	FavColor
0	0	3	female	23.0	7.9250	S	yellow
1	0	1	male	47.0	52.0000	S	purple
2	0	3	male	36.0	7.4958	S	green
3	0	3	male	31.0	7.7500	Q	purple
4	0	3	male	19.0	7.8958	S	purple
...	...	...	...	...	...	...	...

In the Real World...

- ▶ $\mathbb{P}(\text{Survived} = 1) = .408$
- ▶ $\mathbb{P}(\text{Survived} = 1 \mid \text{FavColor} = \text{purple}) = .4$
- ▶ **Not independent...**

In the Real World...

- ▶ $\mathbb{P}(\text{Survived} = 1) = .408$
- ▶ $\mathbb{P}(\text{Survived} = 1 \mid \text{FavColor} = \text{purple}) = .4$
- ▶ **Not independent... ..but “close”!**

In the Real World...

- ▶ $\mathbb{P}(\text{Survived} = 1) = .408$
- ▶ $\mathbb{P}(\text{Survived} = 1 \mid \text{Pclass} = 1) =$

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- ▶ $\mathbb{P}(\text{Survived} = 1) = .408$
- ▶ $\mathbb{P}(\text{Survived} = 1 \mid \text{Pclass} = 1) = .657$

In the Real World...

- ▶ $\mathbb{P}(\text{Survived} = 1) = .408$
- ▶ $\mathbb{P}(\text{Survived} = 1 \mid \text{Pclass} = 1) = .657$
- ▶ **Strong dependence.**

Remember: Conditional Independence

- ▶ Events A and B are **conditionally independent** given C if

$$\mathbb{P}(A, B \mid C) = \mathbb{P}(A \mid C) \cdot \mathbb{P}(B \mid C)$$

- ▶ Equivalently²:

$$\mathbb{P}(A \mid B, C) = \mathbb{P}(A \mid C)$$

²Or $\mathbb{P}(B) = 0$

Informally

- ▶ Suppose you know that C has happened.
- ▶ You have some belief that A happens, given C .
- ▶ A and B are **conditionally independent** given C if learning that B happens in addition to C does not influence your belief that A happens given C .

Very informally

- ▶ A and B are **conditionally independent** given C if learning that B happens in addition to C gives you no more information about A .

$$P(A|C) = \frac{1}{2}$$

$$P(A|B,C) = \frac{1}{2}$$

Example

We've lost the King of Clubs! You draw one card from this deck of 51 cards. A is the event that the card is a heart, B is the event that the card is a face card (J,Q,K,A). Now suppose you know that the card is red. Are A and B independent **given** this information?

Yes! Indep.

♥: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♦: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♣: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, A

♠: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

Titanic Example

- ▶ Survival and class are **not** independent.
 - ▶ $\mathbb{P}(\text{Survived} = 1) = .408$
 - ▶ $\mathbb{P}(\text{Survived} = 1 \mid \text{Pclass} = 1) = .657$
- ▶ But they're (close) to **conditionally independent** given ticket price:
 - ▶ $\mathbb{P}(\text{Survived} = 1 \mid \text{PClass} = 1, \text{Fare} > 50) = .708$
 - ▶ $\mathbb{P}(\text{Survived} = 1 \mid \text{Fare} > 50) = .696$

More Variables

- ▶ $X_1, X_2, \dots, X_d$ are **mutually conditionally independent** given Y if

$$\mathbb{P}(X_1, X_2, \dots, X_d \mid Y) = \mathbb{P}(X_1 \mid Y) \cdot \mathbb{P}(X_2 \mid Y) \cdots \mathbb{P}(X_d \mid Y)$$

Densities

- ▶ If A and B are **continuous** random variables, their joint density can be factored:

$$p(a, b) = p_A(a) \cdot p_B(b)$$

- ▶ If A and B are **conditionally independent** given C , then:

$$p(a, b \mid C = c) = p_A(a \mid C = c) \cdot p_B(b \mid C = c)$$

Densities

- ▶ Suppose $X_1, \dots, X_d$ are d features, Y is class label.
- ▶ If the features are not independent given Y , then:

$$p(\vec{x} \mid Y = y) = p(x_1, x_2, \dots, x_d \mid Y = y)$$

- ▶ **Curse of dimensionality!**

Densities

- ▶ Suppose $X_1, \dots, X_d$ are d features, Y is class label.
- ▶ However, if the features are **mutually conditionally independent** given Y , then:

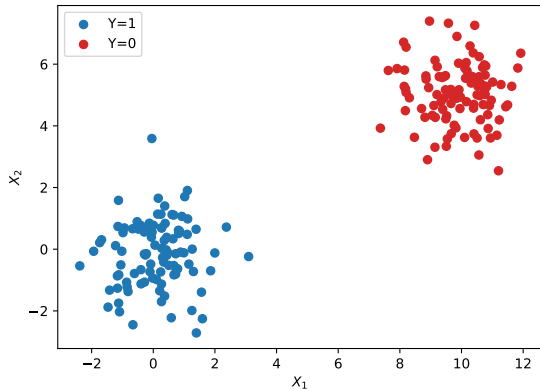
$$\begin{aligned} p(\vec{x} \mid Y = y) &= p(x_1, x_2, \dots, x_d \mid Y = y) \\ &= p_1(x_1 \mid Y = y) \cdot p_2(x_2 \mid Y = y) \cdots p_d(x_d \mid Y = y) \end{aligned}$$

Exercise

Are X_1 and X_2 (close to) conditionally independent given Y ? *Yes.*

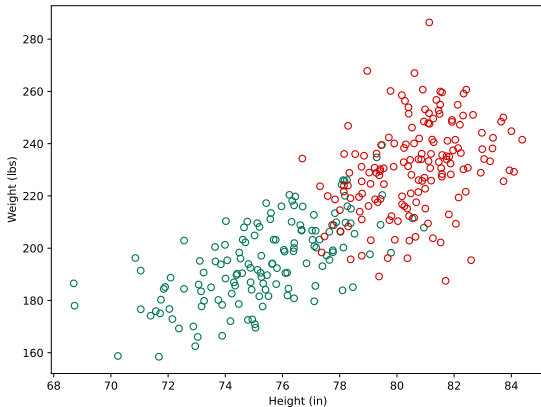
$$p(x_1|x_2, Y=0)$$

$$p(x_1|Y=0)$$



Exercise

Are height and weight (close to) conditionally independent given the player's position? *No!*



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Lecture 15 | Part 3

How Conditional Independence Helps

Recall: The Bayes Classifier

- ▶ To use the Bayes classifier, we must estimate

$$p(\vec{x} \mid Y = y_i)$$

for each class y_i , where $\vec{x} = (x_1, x_2, \dots, x_d)$.

- ▶ Written differently, we need to estimate:

$$p(x_1, \dots, x_d \mid Y = y_i)$$

Recall: Histogram Estimators

- ▶ When $X_1, \dots, X_d$ are continuous, we can use **histogram estimators**.
- ▶ **Curse of Dimensionality**: if we discretize each dimension into 10 bins, there are 10^d bins.

Conditional Independence to the Rescue

- Now suppose $X_1, \dots, X_d$ are mutually conditionally independent given Y . Then:

$$p(x_1, \dots, x_d \mid Y = y_i) = p_1(x_1 \mid Y = y_i) p_2(x_2 \mid Y = y_i) \cdots p_d(x_d \mid Y = y_i)$$

- Instead of estimating $p(x_1, \dots, x_d \mid Y)$, estimate $p_1(x_1 \mid Y), \dots, p_d(x_d \mid Y)$ separately.

Breaking the Curse

- ▶ Suppose we use histogram estimators.
- ▶ If we discretize each dimension into 10 bins, we need:
 - ▶ 10 bins to estimate $p_1(x_1|Y)$
 - ▶ 10 bins to estimate $p_2(x_2|Y)$
 - ▶ ...
 - ▶ 10 bins to estimate $p_d(x_d|Y)$
- ▶ We therefore need $10d$ bins in total.

Breaking the Curse

- ▶ Conditional independence **drastically reduced** the number of bins needed to cover the input space.
- ▶ From $\Theta(10^d)$ to $\Theta(d)$.

Idea

- ▶ Bayes Classifier needs a lot of data when d is big.
- ▶ But if the features are conditionally independent given the label, we don't need so much data.
- ▶ So let's just **assume** conditional independence.
- ▶ The result: the **Naïve Bayes Classifier**.

Naïve Bayes: The Algorithm

- ▶ **Assume** that $X_1, \dots, X_d$ are mutually independent given the class label.
- ▶ Estimate **one-dimensional** densities $p_1(x_1 | Y = y_i), \dots, p_d(x_d | Y = y_i)$ however you'd like.
 - ▶ histograms, fitting univariate Gaussians, etc.
- ▶ Pick the y_i which maximizes

$$p_1(x_1 | Y = y_i) \cdots p_d(x_d | Y = y_i) \mathbb{P}(Y = y_i)$$

But wait...

- ▶ ...are we allowed to just **assume** conditional independence?
- ▶ Sure!
- ▶ The independence assumption is usually **wrong**, but it can work surprisingly well in practice.

Estimating Probabilities

- ▶ You can estimate $p(X_i|Y)$ however makes sense.
- ▶ Popular: **Gaussian Naïve Bayes**.

Example: NBA

- ▶ **Given:** player with height = 75 in, weight = 210 lbs.
- ▶ **Predict:** whether they are a forward or a guard.
- ▶ Let's use Gaussian Naïve Bayes.

Example: NBA

- Compute:

$$p(75 \text{ in}, 210 \text{ lbs} \mid Y = \text{forward})\mathbb{P}(Y = \text{forward})$$

$$p(75 \text{ in}, 210 \text{ lbs} \mid Y = \text{guard})\mathbb{P}(Y = \text{guard})$$

- Using conditional independence assumption:

$$p_1(75 \text{ in} \mid Y = \text{forward}) \cdot p_2(210 \text{ lbs} \mid Y = \text{forward})\mathbb{P}(Y = \text{forward})$$

$$p_1(75 \text{ in} \mid Y = \text{guard}) \cdot p_2(210 \text{ lbs} \mid Y = \text{guard})\mathbb{P}(Y = \text{guard})$$

Example: NBA

- We need to estimate:

$$p_1(75 \text{ in} \mid Y = \text{forward})$$

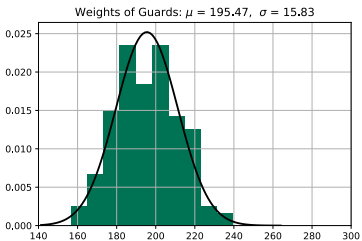
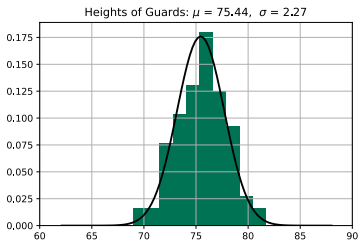
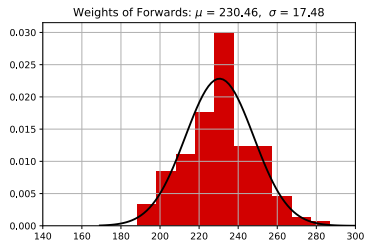
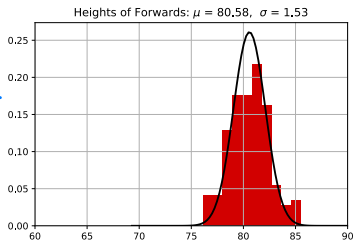
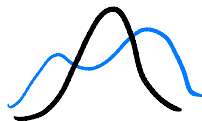
$$p_1(75 \text{ in} \mid Y = \text{guard})$$

$$p_2(210 \text{ lbs} \mid Y = \text{forward})$$

$$p_2(210 \text{ lbs} \mid Y = \text{guard})$$

Example: NBA

- ▶ We'll fit 1-d Gaussians to:
 - ▶ heights of forwards.
 - ▶ heights of guards.
 - ▶ weights of forwards.
 - ▶ weights of guards.



Example: NBA

$$\begin{aligned} & p_1(75 \mid Y = \text{forward}) \cdot p_2(210 \mid Y = \text{forward}) \cdot \mathbb{P}(Y = \text{forward}) \\ &= \mathcal{N}(75; 80.58, 1.53^2) \cdot \mathcal{N}(210; 230.46, 17.48^2) \cdot \frac{156}{300} \\ &\approx 6.73 \times 10^{-6} \end{aligned}$$

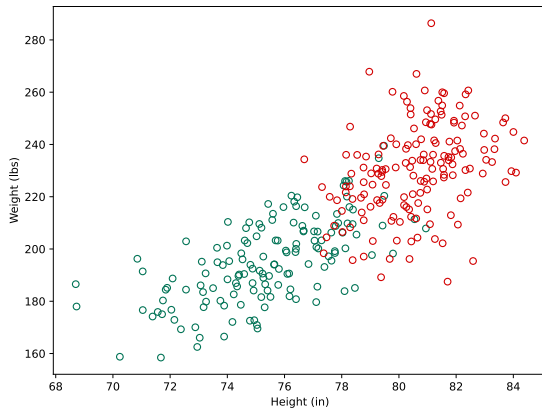
$$\begin{aligned} & p_1(75 \mid Y = \text{guard}) \cdot p_2(210 \mid Y = \text{guard}) \cdot \mathbb{P}(Y = \text{guard}) \\ &= \mathcal{N}(75; 75.44, 2.27^2) \cdot \mathcal{N}(210; 195.47, 15.83^2) \cdot \frac{144}{300} \\ &\approx 5.88 \times 10^{-5} \end{aligned}$$

Example: NBA

- ▶ About 85% accurate on test set.

Exercise

Are height and weight conditionally independent given the player's position?



Example: NBA

- ▶ No!
- ▶ Gaussian Naïve Bayes worked well even though the conditional independence assumption is not accurate.

Gaussian Naïve Bayes

- ▶ $p(X_1 | Y) \cdots p(X_d | Y)$ is a product of 1-d Gaussians with different means, variances.
- ▶ Remember: result is a d -dimensional Gaussian with diagonal covariance matrix:

$$C = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \sigma_d^2 \end{pmatrix}$$

Gaussian Naïve Bayes

- ▶ But in GNB, each class has own diagonal covariance matrix.
- ▶ Therefore: Gaussian Naïve Bayes is **equivalent** to QDA with diagonal covariances.

Beyond Gaussian

- ▶ Naïve Bayes is very flexible.
- ▶ Can use different parametric distributions for different features.
 - ▶ E.g., normal for feature 1, log normal for feature 2, etc.
- ▶ Can use non-parametric density estimation (densities) for other features.
- ▶ Can also handle discrete features.

Up next...

...predicting who survives on the Titanic.

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Probabilistic Modeling & Machine Learning

Lecture 15 | Part 4

The Titanic

The Titanic Dataset

PassengerID	Survived	Pclass	Sex	Age	Fare	Embarked	FavColor
0	0	3	female	23.0	7.9250	S	yellow
1	0	1	male	47.0	52.0000	S	purple
2	0	3	male	36.0	7.4958	S	green
3	0	3	male	31.0	7.7500	Q	purple
4	0	3	male	19.0	7.8958	S	purple
...	...	...	...	...	...	...	...

Goal: predict survival given Age, Sex, Pclass.

Let's use Naïve Bayes

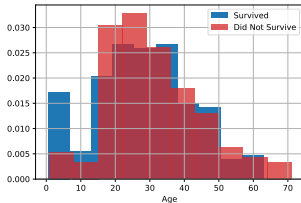
- ▶ We'll pick y_i so as to maximize

$$p(\text{Age} = x_1 \mid Y = y_i) \cdot \mathbb{P}(\text{Sex} = x_2 \mid Y = y_i) \cdot \mathbb{P}(\text{Pclass} = x_3 \mid Y = y_i) \cdot \mathbb{P}(Y = y_i)$$

- ▶ We must choose how to estimate probabilities.
Gaussians?

Estimating Probabilities

- ▶ How do we estimate $p(\text{Age} = x_1 \mid Y = y_i)$?
- ▶ Age is a continuous variable.
- ▶ Looks kind of bell-shaped, we'll fit Gaussians.



Estimating Probabilities

- ▶ How do we estimate $\mathbb{P}(\text{Sex} = x_1 \mid Y = y_i)$?
- ▶ Sex is a **discrete** variable in this data set.
- ▶ Fitting Gaussian makes no sense.
- ▶ But estimating these probabilities is easy.

Estimating Probabilities

$$\mathbb{P}(\text{Sex} = \text{male} \mid \text{Survived}) \approx \frac{\# \text{ of survived and male}}{\# \text{ of survived}} \\ = .4$$

$$\mathbb{P}(\text{Sex} = \text{male} \mid \text{Did Not Survive}) \approx \frac{\# \text{ of died and male}}{\# \text{ of died}} \\ = .87$$

Estimating Probabilities

- ▶ Pclass, too, is categorical. Estimate in same way.
- ▶ You can estimate $\mathbb{P}(X_i|Y)$ however makes sense.
- ▶ **Can use different ways for different features.**
- ▶ Gaussian for age, simple ratio of counts for class, sex.

Example: The Titanic

- ▶ Using just age, sex, ticket class, Naïve Bayes is 70% accurate on test set.
- ▶ Not bad. Not great.
- ▶ To do better, add more features.

In High Dimensions

- ▶ Naïve Bayes can work well in high dimensions.
- ▶ Example: document classification.
 - ▶ Document represented by a “bag of words”.
 - ▶ Pick a large number of words; say, 20,000.
 - ▶ Make a d -dimensional vector with i th entry counting number of occurrences of i th word.

Practical Issues

- ▶ We are multiplying lots of small probabilities:

$$\mathbb{P}(X_1 | Y) \cdots \mathbb{P}(X_d | Y)$$

- ▶ Potential for **underflow**.

Practical Issues

- ▶ “Trick”: work with log-probabilities instead.
- ▶ Pick the y_i which maximizes

$$\begin{aligned} & \log [\mathbb{P}(X_1 = x_1 \mid Y = y_i) \cdots \mathbb{P}(X_d = x_d \mid Y = y_i) \mathbb{P}(Y = y_i)] \\ &= \log \mathbb{P}(X_1 = x_1 \mid Y = y_i) + \dots + \log \mathbb{P}(X_d = x_d \mid Y = y_i) + \log \mathbb{P}(Y = y_i) \\ &= \left(\sum_{j=1}^d \log \mathbb{P}(X_j = x_j \mid Y = y_i) \right) + \log \mathbb{P}(Y = y_i) \end{aligned}$$